

AMALIA



December 24, 1992



amaliahwang@gmail.com



010-5700-1224



<https://scholar.google.com/citations?user=EV-yPOQAAAAJ&hl=en>



<https://orcid.org/0009-0005-9828-5431>



Postdoctoral researcher with experience in wireless communications, focusing on mobile wireless networks, clustering, QoS routing & multicast routing, network layer security, and machine learning applications. Also experienced in lecturing and teaching artificial intelligence-related courses.

Academic Background

- 2022 – 2026  **Ph.D., GPA 4.29 out of 4.50**
Department of Software and Communications Engineering, Hongik University, South Korea
Thesis title: *QoS-Aware Multicast Routing Protocols for Future Mobile Wireless Networks: A Deep Learning Approach*
- 2016 – 2018  **Master of Engineering, GPA 3.90 out of 4.00**
Department of Electrical Engineering, Politeknik Negeri Malang, Indonesia
Thesis title: *Numerical Analysis of IFA for UHF and SHF Channel Systems*
- 2015 – 2016  **Bachelor of Applied Science, GPA 3.77 out of 4.00**
Department of Informatics Engineering, Politeknik Negeri Malang, Indonesia
Thesis title: *Development of a Home Ownership Credit Eligibility System Using the SAW (Simple Additive Weighting) Method (Case Study: PT. Bank Syariah Bukopin, Sidoarjo Branch)*
- 2010 – 2013  **Associate Degree of Applied Science, GPA 3.87 out of 4.00**
Department of Computer Engineering, Telkom University, Bandung, Indonesia
Thesis title: *Analysis and Implementation of Quality of Service (QoS) on a MAN Network Using a Huawei Router (Case Study: Telkom RDC Gegerkalong, Bandung)*

Work Experience

- March 2026 – Now  **Postdoc Researcher**
AI Security Convergence Lab, Hankyong National University, Gyeonggi-do, Republic of Korea
- 2019 – 2022  **Lecturer**
Department of Electrical Engineering, Politeknik Negeri Malang, Indonesia
Assigned to teach several subjects such as Antenna, Artificial Intelligence, Modulation System, Broadband, Transmission Line, and Network Security System
- 2014 – 2015  **IT Staff**
Syariah Bukopin Bank, Surabaya, Indonesia
- March – June 2013  **Internship**
Transmission Lab at PT. Telkom RDC Gegerkalong, Bandung, Indonesia
Configuring QoS with DSCP and 802.1p marking on Huawei Router

Research Publications

Journal Articles

- 1 **A. Amalia**, Y. Pramitarini, R. Hendra Yoga Perdana, K. Shim, and B. An, "A Cross-Layer FL-Based Clustering Protocol to Support Multicast Routing in IoT-Enabled MANETs With CF-mMIMO," *IEEE Access*, vol. 13, pp. 3881–3899, 2025.

- 2 A. **Amalia**, Y. Pramitarini, R. H. Y. Perdana, K. Shim, and B. An, "Beamforming-Based Multicast Routing Protocol in Underlay Cognitive MANETs with STAR-RIS: Deep Learning Design," *IEEE Internet Things Journal*, 2025, (Submitted).
- 3 A. **Amalia**, Y. Pramitarini, R. H. Y. Perdana, K. Shim, and B. An, "Energy Efficient Multicast Routing Protocol Using FL-Based Optimal Route Selection in IoT-Enabled MANETs With RIS and CF-mMIMO," *IEEE Internet Things Journal*, vol. 12, no. 22, pp. 47 588–47 606, 2025.
- 4 A. **Amalia**, Y. Pramitarini, R. H. Y. Perdana, K. Shim, and B. An, "A Deep-Learning-Based Secure Routing Protocol to Avoid Blackhole Attacks in VANETs," *Sensors*, vol. 23, no. 19, 2023, ISSN: 1424-8220.
- 5 E. Rohadi, M. F. Ali, I. Siradjuddin, A. Setiawan, A. **Amalia**, and Y. Darmawan, "Design and Analysis of Ultra Low-Profile ILA on a Rectangular Conducting Plane," *International Journal of Engineering & Technology*, vol. 7, no. 4.44, pp. 55–60, 2018.
- 6 E. Rohadi, A. **Amalia**, I. Siradjuddin, *et al.*, "Design and Analysis of The IFA Bandwidth Enhancement for 639 MHz UHF Channel," *International Journal of Engineering & Technology*, vol. 7, no. 4.44, pp. 61–64, 2018.

Conference Proceedings

- 1 A. **Amalia**, Y. Pramitarini, Y. Perdana Ridho Hendra, K. Shim, and B. An, "Federated Learning-Based Clustering Protocol Utilizing Cross-Layer Design in IoT-Enabled MANETs with CF-mMIMO," in *The 13th International Conference on Green and Human Information Technology (ICGHIT)*, Nha Trang, Vietnam, 2025, pp. 52–54.
- 2 A. **Amalia**, Y. Pramitarini, R. H. Y. Perdana, K. Shim, and B. An, "Federated Learning-Enhanced QoS Multicast Routing to Support RIS and Edge Computing in IoT-Enabled MANETs with CF-mMIMO," in *2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC)*, Okinawa, Japan, 2024, pp. 1–6.
- 3 A. **Amalia**, Y. Pramitarini, Y. Perdana Ridho Hendra, K. Shim, and B. An, "Federated Learning-Based Clustering Protocol to Support QoS Routing in Vehicle Edge Computing," in *The 12th International Conference on Green and Human Information Technology (ICGHIT)*, Hanoi, Vietnam, 2024, pp. 168–173, ISBN: 2013206534.
- 4 A. **Amalia**, Y. Pramitarini, Y. Perdana Ridho Hendra, K. Shim, and B. An, "Cuckoo Optimization-Based Routing Protocol to Enhance QoS in Mobile Ad-Hoc Networks," in *The 11th International Conference on Green and Human Information Technology (ICGHIT)*, Bangkok, Thailand, 2023, pp. 244–249.
- 5 E. Rohadi, A. **Amalia**, D. K. Kusumah, G. A. A. Fahrezi, W. I. Sabilla, and I. Siradjuddin, "The Development and Analysis of An Information System for Incentives Publication of Reputable Indexed International Journal and National Journal (Case Study State Polytechnic of Malang)," in *IOP Conference Series: Materials Science and Engineering*, vol. 1073, IOP Publishing, 2021, p. 012 056.
- 6 E. Rohadi, A. **Amalia**, R. F. D. Qoyyimah, R. Ardiansyah, and I. Siradjuddin, "Study on Information System Infrastructure in Probolinggo by using Self Organizing Map," in *IOP Conference Series: Materials Science and Engineering*, vol. 1098, IOP Publishing, 2021, p. 052 106.
- 7 E. Rohadi, C. Rahmad, F. Chrissandy, and A. **Amalia**, "Study on Network Management Systems by using Docker Kubernetes," in *IOP Conference Series: Materials Science and Engineering*, vol. 1098, IOP Publishing, 2021, p. 022 093.
- 8 E. Rohadi, A. **Amalia**, J. D. Bagaskara, B. Harijanto, and S. Adhisuwarnjo, "GIS for Coffee Shops Classification and Routing using Naive Bayes Method," in *IOP Conference Series: Materials Science and Engineering*, vol. 732, IOP Publishing, 2020, p. 012 079.
- 9 E. Rohadi, A. **Amalia**, A. Prasetyo, M. F. Rahmat, A. Setiawan, and I. Siradjuddin, "Cluster Implementation on Mini Raspberry Pi Computers using Round Robin Algorithm," in *Journal of Physics: Conference Series*, vol. 1450, IOP Publishing, 2020, p. 012 068.

- 10 E. Rohadi, M. Sarosa, M. F. Ali, **A. Amalia**, A. Setyawan, and S. Adhisuwarnjo, "Design and Analysis of Planar Wideband Monopole Antenna using WIPL-D Simulator," in *IOP Conference Series: Materials Science and Engineering*, vol. 732, IOP Publishing, 2020, p. 012 100.
- 11 E. Rohadi, A. Adzikirani, I. Siradjuddin, *et al.*, "The Study of The Wideband Planar Sleeve Antenna," in *Journal of Physics: Conference Series*, vol. 1402, IOP Publishing, 2019, p. 033 099.
- 12 E. Rohadi, **A. Amalia**, A. Nidianingsih, S. N. Arief, R. Ariyanto, and D. W. Wibowo, "The Design and Evaluation of An Automatic Watering System by using Fuzzy Mamdani," in *Journal of Physics: Conference Series*, vol. 1402, IOP Publishing, 2019, p. 022 088.
- 13 E. Rohadi, **A. Amalia**, I. Siradjuddin, M. Taguchi, A. Setiawan, and S. Adhisuwarnjo, "Study of Parasitic Element Effects of Multiband IFA for UHF and SHF Channel Systems," in *2019 Conference on Antenna Measurements & Applications (CAMA)*, 2019, pp. 184–187.
- 14 E. Rohadi, S. A. Suwarnjo, M. C. Pradana, and **A. Amalia**, "Internet of Things: CCTV Monitoring by Using Raspberry Pi," in *2018 International Conference on Applied Science and Technology (iCAST)*, 2018, pp. 454–457.
- 15 C. Setiadi, E. Rohadi, M. Sarosa, **A. Amalia**, and A. Setiawan, "Numerical analysis and design of inverted L antenna for UHF TV receiver application," in *2017 International Conference on Sustainable Information Engineering and Technology (SIET)*, 2017, pp. 471–474.

Skills

Languages	📖 Bahasa Indonesia, English, Korean
Coding Languages	📖 C/C++, Python, MATLAB, JAVA, PHP
Simulators	📖 NS3, Packet Tracer, WIPL-D
Deep Learning Tools	📖 VS Code, Spyder, MATLAB
Web Programming Tools	📖 VS Code, NetBeans, Eclipse IDE
UI/UX Programming Tools	📖 Android Studio, Visual Studio

Miscellaneous Experience

Scholarship

2022-2026	📖 Global Scholarship for PhD Program from Hongik University, South Korea
2016-2018	📖 Scholarship for Master Program from Politeknik Negeri Malang, Indonesia

Certification

2015	📖 Cisco Certified Network Associate (CCNA 1-4)
2013	📖 Juniper Networks Certified Associate (JNCIA-Junos)